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Question: Should we report posterior mean, median or mode?
 (2-4 min) Which is best?
 What does it mean for an estimator to be "best"?

Answer: Depends on loss function.

We can view Bayes estimators as "risk minimizers"

Setup

parameters θ

Data y

Loss function $L(\theta, a)$ - determines the cost of deciding that the parameter has value a when it is in fact equal to θ .

The Bayes estimator, $\hat{\theta}_B$ minimizes posterior expected loss.

Example: $L(\theta, a) = (\theta - a)^2$

Posterior Risk: $R(a|y) = \underbrace{\int (\theta - a)^2 p(\theta|y) d\theta}_{\mathbb{E}[L(\theta, a) | y]}$

$\hat{\theta}_B := \underset{a}{\operatorname{argmin}} \mathbb{E}[(\theta - a)^2 | y]$, expand out:
 $= \underset{a}{\operatorname{argmin}} \mathbb{E}[\theta^2 | y] - 2\mathbb{E}[\theta \cdot a | y] + \mathbb{E}[a^2 | y]$
 * expectation taken w.r.t $\theta | y$. $\Rightarrow$
 $= \underset{a}{\operatorname{argmin}} \underbrace{\mathbb{E}[\theta^2 | y]}_{\text{constant in } a} - 2a \mathbb{E}[\theta | y] + a^2$

* differentiate w.r.t. a ; set = 0:

$$-2\mathbb{E}[\theta | y] = -2a \Rightarrow \boxed{a = \mathbb{E}[\theta | y]}$$

(2)

Ex Absolute error loss

$$L(\theta, a) = |\theta - a|$$

$$\text{Risk}(a|y) = \mathbb{E}[|\theta - a| | y] = \int_{-\infty}^{\infty} |\theta - a| p(\theta|y) d\theta$$

* Split integral at a :

$$= \int_{-\infty}^a (a - \theta) p(\theta|y) d\theta + \int_a^{\infty} (\theta - a) p(\theta|y) d\theta$$

* Differentiate w.r.t. a :

$$\frac{d}{da} \text{Risk}(a|y) = \int_{-\infty}^a p(\theta|y) d\theta - \int_a^{\infty} p(\theta|y) d\theta$$

* set = 0:

$$\int_{-\infty}^a p(\theta|y) d\theta = \int_a^{\infty} p(\theta|y) d\theta$$

$$\Rightarrow P(\theta \leq a|y) = \frac{1}{2}$$

$$\Rightarrow \boxed{\hat{\theta}_B := \text{posterior median}}$$

Contrast w/ frequentist estimators:

$$\begin{aligned} \text{Frequentist risk: } R(\theta, \delta) &= \mathbb{E}_Y[L(\theta, \delta(Y))] \\ &= \int L(\theta, \delta(y)) \cdot p(y|\theta) dy \end{aligned}$$

$\nwarrow$ fixed